Cross-Domain Hyperspectral Image Classification via Mamba–CNN and Knowledge Distillation

Aoyan Du, Guixin Zhao, Mengxin Cao, *Member, IEEE*, Yuyao Wang, Aimei Dong, Guohua Lv, Yongbiao Gao, Dongqing Li, and Xiangjun Dong

***Abstract*—Domain adaptation (DA)-based cross-domain hyperspectral image (HSI) classification methods have garnered significant attention. The majority of DA techniques utilize models based on convolutional neural networks (CNNs) and Transformers for feature extraction. However, Transformers may struggle to capture local details in HSIs, while CNNs often underperform in handling long-range dependencies. Furthermore, many methods focus only on aligning marginal distributions while ignoring the consistency of interclass features, which may lead to feature confusion and degraded classification accuracy. To overcome the challenges mentioned, we propose a Mamba–CNN and knowledge distillation network (MKDnet). First, the network employs a feature extractor that integrates Mamba and CNN frameworks for cross-domain HSI classification, enabling the capture of both global and local features while effectively capturing long-range dependencies. Second, domain alignment is achieved through distribution alignment and graph alignment. In the distribution alignment phase, we design a knowledge distillation architecture that utilizes soft labels to enhance the understanding of relationships between classes, thereby improving the consistency of interclass features. In the graph alignment phase, we use graph convolution to capture connections between nodes and edges and transfer class-level topological relationships across domains. Finally, the classifier is used to obtain classification results, with consistency constraints applied to balance features between classes more effectively. Extensive experiments have demonstrated that MKDnet outperforms other state-of-the-art methods on three public cross-domain HSI datasets.**

***Index Terms*—Cross-domain, domain adaptation (DA), hyperspectral image (HSI) classification, knowledge distillation, Mamba.**

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Aoyan Du, Guixin Zhao, Mengxin Cao, Aimei Dong, Guohua Lv, Yongbiao Gao, and Xiangjun Dong are with the Key Laboratory of Computing Power Network and Information Security, Ministry of Education, Shandong Computer Science Center (National Supercomputer Center in Jinan), Qilu University of Technology (Shandong Academy of Sciences), Jinan 250316, China, also with the Faculty of Computer Science and Technology, Qilu University of Technology (Shandong Academy of Sciences), Jinan 250316, China, and also with Shandong Provincial Key Laboratory of Computing Power lnternet and Service Computing, Shandong Fundamental Research Center for Computer Science, Jinan 250014, China (e-mail: zgx@qlu.edu.cn; dxj@qlu.edu.cn).

Yuyao Wang is with the Department of Mathematics and Statistics, Boston University, Boston, MA 02215 USA (e-mail: yuyaow@bu.edu).

Dongqing Li is with the Systems Engineering Research Institute, China State Shipbuilding Corporation Ltd., Beijing 100094, China (e-mail: lidongqingcumt@163.com).

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I. Introduction

RECENTLY, the application of various remote sensing data, including hyperspectral image (HSI) [1], light detection and ranging (LiDAR) [2], and synthetic aperture radar (SAR) [3], has become a key focus in land cover classification research. These advanced imaging techniques offer diverse data sources that enable a more comprehensive analysis of land surface characteristics. Compared to multispectral images [4], HSI [5] provides significantly richer spectral information by capturing data across a wide range of wavelengths. This detailed spectral information enables more accurate identification of land cover materials and categories, allowing for better differentiation. However, most HSI classification methods are applicable only to fixed scenes and cannot be easily adapted to other environments [6], [7]. Cross-domain HSI classification aims to address this challenge [8], [9]. In this context, source data (SD) refers to the training samples, while target data (TD) refers to the test samples. The main goal is to use SD to train the model for precise TD classification in the target scene.

The domain shift refers to the difference between SD and TD, which can reduce the performance of machine learning models across different domains [10], [11]. In the HSI classification, domain shift presents a significant challenge for models to maintain consistent classification accuracy, particularly when models trained on the SD are directly applied to classify data in the TD. The HSI acquisition process is highly sensitive to several factors, including sensor nonlinearities [12], varying weather conditions [13], and seasonal changes [14]. These factors can cause significant variations in spectral reflectance for the same land cover class across different domains, which exacerbates the domain shift problem and adversely affects model performance. To tackle these challenges, transfer learning has emerged as an effective solution and has been widely applied in cross-domain HSI classification [15], [16]. As a particular case of transfer learning, domain adaptation (DA) assumes that the tasks in the SD and TD are identical, even though the domains are distinct yet related. DA's objective is to minimize the effects of domain shift. [17], [18].

In cross-domain HSI classification tasks, traditional DA approaches are categorized into three main types: distribution alignment methods [19], [20], [21], feature selection methods [22], [23], [24], and subspace learning methods [25], [26], [27]. Distribution alignment methods mitigate domain shift by aligning data distributions between the SD and TD [28]. Feature selection methods focus on identifying common features in both domains to build a shared feature space

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[29]. Subspace learning methods perform DA by minimizing distribution discrepancies between SD and TD when projected into a common subspace [30]. While these approaches address domain shift to some extent, they often overlook cases where the distributions of SD and TD may vary significantly. This limitation is particularly evident when the edge distributions of SD and TD exhibit marked differences.

Compared to traditional DA methods, recent deep learning (DL) techniques have introduced adaptive layers aimed at minimizing the domain shift, thereby improving classification performance on the TD. For instance, the deep adaptation network (DAN) [31] addresses domain shift by applying the multikernel maximum mean discrepancy (MK-MMD) metric across its adaptive layers. The multirepresentation adaptation network (MRAN) [32] tackles cross-domain classification by aligning various feature representations, effectively utilizing diverse information to measure class-conditional distribution differences. Furthermore, Zhang et al. [33] introduced the topology structure and semantic information transfer network (TSTnet), which leverages graph convolution to capture topological relationships in HSI classification.

The success of DA methods relies heavily on effective feature extraction. In HSI tasks, it is crucial to capture both local and global features because of the data's high-dimensional complexity. Convolutional neural networks (CNNs) [34] are commonly employed to capture local features [35]. This is particularly useful for capturing fine details. However, CNNs are limited in capturing global features, which are essential for understanding long-range dependencies in HSI classification. Transformers [36] excel at capturing global context and long-range dependencies through self-attention mechanisms, making them effective for cross-domain HSI classification. However, Transformers [37] suffer from quadratic computational complexity as input size increases, which can be a limitation for large HSI datasets. Mamba, a new state space model (SSM) [38], [39], effectively captures long-range dependencies, thus overcoming the computational burden of Transformers. Mamba is capable of efficiently capturing global features, which is essential for accurate HSI classification.

Despite significant advancements in cross-domain HSI classification, challenges exist in feature extraction and domain alignment.

1) Some approaches primarily emphasize marginal distribution alignment between the SD and TD. However, they often overlook the critical importance of maintaining consistency in interclass features. This oversight can reduce classification performance by causing confusion between classes. Consequently, this shortcoming affects the effectiveness of domain alignment [40], [41]. As a result, domain shift remains a major challenge for improving cross-domain HSI classification performance [42].
2) General approaches often use CNNs [43], [44] for local feature extraction. Although CNNs capture local details well, they struggle with long-range dependencies. In contrast, Transformer models [45], [46] effectively capture contextual information but often overlook intricate details in cross-domain HSI classification, affecting feature extraction quality. While Mamba [47] handles long-range dependencies well, it does not specifically address local features. Thus, comprehensively extracting features remains a critical issue.

To tackle the aforementioned challenges, we propose a Mamba–CNN and knowledge distillation network (MKDnet). This network uses a feature extractor that combines both Mamba and CNN architectures. Compared to other methods, MKDnet excels at capturing both global and local features while efficiently handling long-range dependencies. During the distribution alignment phase, we introduce a knowledge distillation architecture (STKD) that uses soft labels to enhance interclass feature consistency. To explore the topological relationships in HSI, we employ a graph convolutional network that dynamically updates subgraph information, optimizing the graph alignment process. Moreover, both distribution alignment and graph alignment are key components of domain alignment, applying consistency constraints to their outputs, thus mitigating the effects of domain shift.

The main contributions of this article are summarized as follows.

1) We propose MKDNet, a cross-domain HSI classification framework with a dual-branch extractor combining Mamba and CNN to capture local and long-range features. Through two-level domain alignment, MKDNet improves class discriminability and classification accuracy.
2) We design a knowledge distillation-based alignment module that performs probability-based alignment to improve class discriminability and graph-based alignment to enhance cross-domain consistency.
3) We impose consistency constraints between distribution and graph alignment outputs, promoting agreement between two feature views. This dual-view alignment reduces feature imbalance and improves training stability across domains.

The structure of this article is organized as follows. Section II provides a brief overview of the related work. Section III presents a detailed explanation of the principles behind the proposed MKDnet method. In Section IV, we demonstrate the effectiveness and advantages of the proposed method through comprehensive experiments and analysis. Finally, Section V concludes this article with a summary of our work.

II. Related Work

A. Mamba

Mamba is a novel SSM proposed by Gu and Dao et al. [39]. This model enhances performance by introducing a selective state space, enabling it to selectively propagate information according to the current token. Recent advancements in Mamba [48] have broadened its applicability to various tasks, offering a new approach to modeling complex dependencies in high-dimensional data. These models integrate classical control theory with modern DL techniques, providing a unique capability to capture intricate patterns in large datasets. While

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Mamba has proven effective for low-dimensional data such as audio and speech, its potential for high-dimensional visual data, particularly HSI, remains largely underexplored [49], [50]. HSI contains rich spectral information across various wavelengths and is essential for applications that require precise analysis of materials and environmental conditions. This article aims to explore the application of Mamba in processing HSI data, harnessing its strengths to handle the complexities of high-dimensional information. By incorporating the unique characteristics of Mamba, we seek to enhance data interpretation and feature extraction in HSI analysis.

B. Knowledge Distillation

As DL models increase in size and complexity, so does the demand for computational resources. To address this challenge, Hinton et al. [51] designed knowledge distillation—a model compression technique where a student model learns not only from the true labels in the dataset but also from the soft labels produced by a teacher model during training. This process helps capture finer interclass features. Soft labels are generally obtained from the predicted probability distribution of the teacher model for each class. Compared to hard labels (such as binary classification results of 0 or 1), soft labels offer a more nuanced representation of the relationships and similarities between different classes. This approach has spurred numerous innovations, enabling student models to achieve performance comparable to that of their teacher models, making them particularly well-suited for deployment in resource-constrained environments [52]. While knowledge distillation has been successfully applied in fields like natural language processing [53] and image recognition [54], its application in cross-domain HSI classification remains limited. The rich spectral and spatial features of HSI data pose challenges for traditional distillation methods, which often focus on lower-dimensional data. Moreover, the high dimensionality of HSI data complicates knowledge transfer between the SD and TD due to varying data distributions.

C. Graph Neural Network

Graph neural networks (GNNs) were initially designed for tasks involving graph-structured data [55]. They primarily represent the features of a central node by aggregating information from its neighborhood and recursively transforming the features of neighboring nodes.

The framework of GNNs recursively aggregates to update the features of neighboring nodes, making each node's representation dependent on both its own features and the structure and features of its neighbors. Through multiple layers, GNNs capture increasingly broader graph structural information. Many GNN methods, like graph sampling aggregation network (GraphSAGE) [56], approach this from the perspective of set functions. As illustrated in Fig. 1, in GraphSAGE, nodes iteratively gather information from their local neighbors, gradually acquiring insights from further parts of the graph.

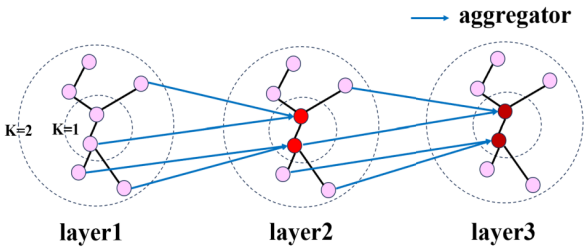


Fig. 1. Diagram of the GraphSAGE.

III. Proposed Method

The MKDnet proposed in this article, as shown in Fig. 2, is comprised of three key components: feature extractor, domain alignment (including distribution alignment and graph alignment), and classifiers. Specifically, we randomly select labeled patches from the SD and unlabeled patches from the TD, then input them into both global and local Mamba–CNN (GLMC) feature extractors. The GLMC module comprises two branches: one extracts global features using Mamba, while the other extracts local features with a 2DCNN. Together, these branches provide a richer set of features, with the features from SD and TD represented as M_s and M_t, respectively. M_s and M_t are fed into the domain alignment framework. In the distribution alignment process, knowledge distillation techniques are employed to ensure consistency of interclass features. For graph alignment, the graph features extracted by GraphSAGE from M_s and M_t are denoted as G_s and G_t, respectively, and the features of each layer are aligned using the graph optimal transport (GOT) method. Finally, probability predictions are made using M_s and G_s, ensuring distribution consistency to achieve improved interclass features balance. During the final testing phase, TD is classified using a CNN designed to extract domain-invariant features.

A. Feature Extractor

To more effectively extract both global and local features from HSI, we use a feature extractor called GLMC, as illustrated in Fig. 3. GLMC is a dual-branch module. First, the SD and TD are split into two parts along the channel dimension, with each part being input into its corresponding branch: the Mamba branch and the Conv branch.

The input is first processed through layer normalization, after which it is split into two subbranches in the Mamba branch. The first subbranch consists of a linear layer followed by an activation function. The input is first processed through a linear layer and a convolution and then passed into a 2-D selective scan module (SS2D)—a 2-D spatial traversal mechanism with four scanning paths that unfolds input blocks into sequences along four distinct traversal routes. This allows each pixel in the image to integrate information from other pixels in different directions, helping establish a global receptive field in 2-D space—followed by layer normalization for further feature extraction in the second subbranch. After normalization, the outputs from the two subbranches are multiplied element-wise and merged. Finally, a linear layer performs feature mixing,

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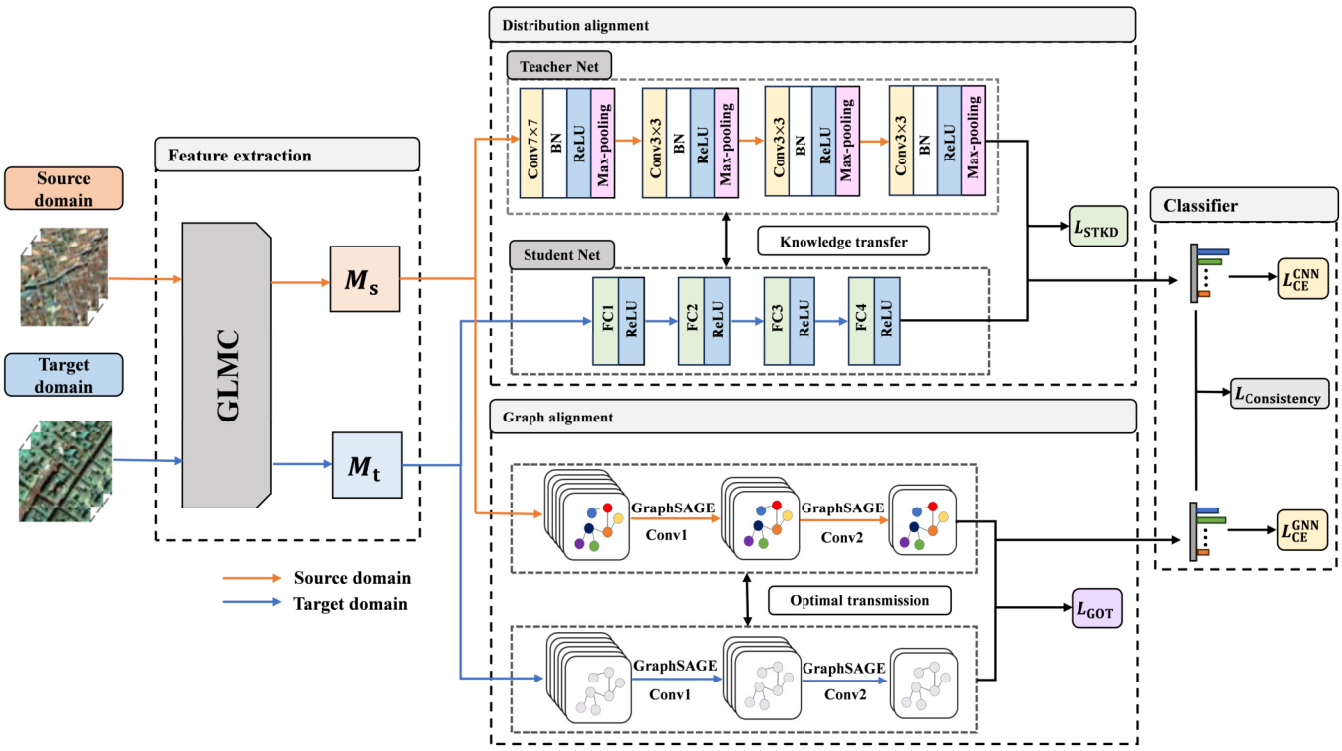


Fig. 2. Diagram of the proposed MKDnet.

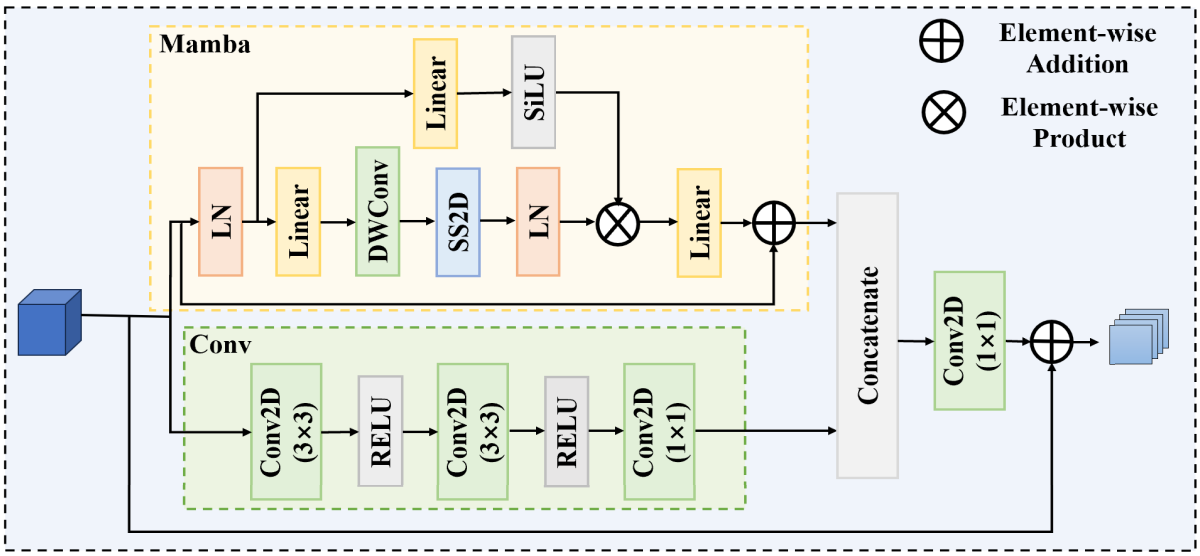


Fig. 3. Diagram of the proposed GLMC.

and a residual connection is applied to generate the output of the Mamba branch. It is worth noting that the activation function in the Mamba branch is set to SiLU by default. In the Conv branch, three convolutional layers (Conv2D 1×1, Conv2D 3×3, and Conv2D 3×3) are employed to capture the input's local features.

At the end of the network, the outputs of the two branches are concatenated along the channel dimension. A 1×1 convolution layer is then applied to fuse the feature maps, allowing for channel interaction. Finally, the extracted feature M_s and M_t is obtained. This process effectively captures both global and local features, leading to a more comprehensive feature extraction mechanism.

B. Domain Alignment

1) Ditribution Alignment: To tackle the issue of domain shift in cross-domain HSI classification, we design a knowledge distillation-based distribution alignment framework that

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includes two key components: a teacher network and a student network. Both networks have a four-layer architecture, with the student network being trained to approximate the predictive distribution of the teacher network. The specific design and distribution alignment structure are shown in Fig. 2.

First, M_s and M_t are input into the distribution alignment framework. The teacher network, utilizing the richer feature information in the SD features, generates a more comprehensive representation in the form of soft labels, which encapsulate the probability distribution across different classes. This rich representation not only conveys critical interclass features but also enhances the overall learning process. We adopt a Conv 7×7 in the first layer of the teacher network to capture broader spatial features, which helps extract coarse-grained spatial structure. The subsequent layers use Conv 3×3 to gradually refine local features while maintaining computational efficiency. Although the student network processes the TD features separately, it learns to align its output with the more detailed label information from the teacher network through knowledge distillation. Since the teacher network is trained on more informative SD features, its output is typically more accurate. By transferring this richer label information to the student network, it captures higher level patterns and gains a deeper understanding of the TD features. This transfer of knowledge effectively addresses the challenges posed by domain shift, resulting in improved classification performance and greater model robustness. Overall, this process reduces the performance gap between the two networks, facilitating better classification by aligning the student network's output with that of the teacher network.

During the knowledge transfer from the teacher network to the student network, the resulting loss is referred to as the distillation loss L_{distill}. The detailed formula is as follows:

$$L_{\text{distill}} = \text{KL}\left(S\left(\frac{z^s}{T}\right),\ S\left(\frac{z^t}{T}\right)\right) \times T^2 \tag{1}$$

where KL represents the KL divergence, $S(\bullet)$ is the softmax function with temperature, z^s is the output of the teacher network, T is the temperature coefficient, and z^t is the output of the student network. By aligning these two features, we can ensure that the SD and TD features have similar spatial representations, thus mitigating cross-domain distributional differences. The loss generated in this process is referred to as the alignment loss, and the detailed formula is as follows:

$$L_{\text{align}} = \frac{1}{N}\sum_{i=1}^{N}\left|\left|z_s^i - z_t^i\right|\right|_2^2 \tag{2}$$

where N represents the number of features and $\| \|_2$ denotes the second normal form.

The final loss function is the weighted total of the above losses and is specifically expressed as

$$L_{\text{STKD}} = \alpha \cdot L_{\text{distill}} + (1-\alpha) \cdot L_{\text{align}} \tag{3}$$

where α is a balancing parameter, and we set it to 0.5 for equal weighting between the two loss terms.

2) Graph Alignment: The framework employs a two-layer GraphSAGE based on a mean aggregation function and dynamically constructs deep feature subgraphs for graph alignment. The formula is for the GraphSAGE algorithm

$$G_v^l \leftarrow \sigma\left(W^{l-1} \cdot \text{Mean}\left(\left\{G_v^{l-1}\right\} \cup \left\{G_u^{l-1}\ \forall u \in N(v)\right\}\right)\right) \tag{4}$$

where v represents the central node, $\sigma(\bullet)$ is a nonlinear activation function, W^{l-1} is the weight matrix of the first layer, and Mean$(\bullet)$ represents the aggregation function between the central node and selected neighboring nodes. u represents its neighboring node. $N(v)$ denotes the neighborhood function controlling the order of neighboring node selection. In addition, the GOT distance is defined to measure the similarity between subgraphs

$$\text{GOT}(G_s, G_t) = \sum_{i,\hat{i},j,\hat{j}}\Big(T_{ij}\gamma \cdot c\left(g_i^s \cdot g_j^t\right) + (1-\gamma) \cdot T_{\hat{i}\hat{j}} L\left(g_i^s, g_j^t, \widehat{g_l^s}, \widehat{g_l^t}\right)\Big) \tag{5}$$

where T_{ij} and $T_{\hat{i}\hat{j}}$ are the transformation matrices for two pairs of nodes [i.e., g_i^s and g_j^t and $\widehat{g_i^s}$ and $\widehat{g_j^t}$]. In the experiment, γ is set to 0.5. Therefore, the GOT loss can be expressed as

$$L_{\text{GOT}}(M_s, M_t) = \sum_{l=0}^{2} GOT\left(G_s^l, G_t^l\right) \tag{6}$$

where G is the output of the first layer of the GNN. The notation $l = 0$ denotes the original subgraph features generated by M_s and M_t, while $l = 1$ and $l = 2$ correspond to the graph convolution features from the first and second layers, respectively.

C. Classifiers of MKDnet

In MKDnet, CNN and GNN are used as classifiers to perform classification tasks in SD, with their classification losses and prediction probability calculations defined as follows:

$$L_{\text{CE}}^{\text{CNN}}(X_s, Y_s) = \frac{1}{n_s}\sum_{i=1}^{n_s} L_y\left(S\left(z_i^s\right), y_i\right) \tag{7}$$

$$L_{\text{CE}}^{\text{GNN}}(M_s, Y_s) = \frac{1}{n_s}\sum_{i=1}^{n_s} L_y\left(S\left(g_i^s\right), y_i\right) \tag{8}$$

where $S(\bullet)$ is the softmax function and X_s and Y_s represent the data and labels from the SD, respectively.

The cross-entropy loss function is used to quantify the difference between the model's predicted probability distribution and the actual label distribution. The model is trained to align its output probability distribution more closely with the true distribution by minimizing this loss. These formulas ensure that both graph-structured features and image features are effectively used for classification, thus improving the overall performance of MKDnet. $L_{\text{Consistency}}(\bullet)$ is the constraint on the probability predictions of $L_{\text{CE}}^{\text{CNN}}$ and $L_{\text{CE}}^{\text{GNN}}$ for distribution consistency

$$L_{\text{Consistency}}(M_s, G_s) = -\frac{1}{n_s}\sum_{i=1}^{n_s}\sum_{c=1}^{C} p_i^c \log q_i^c \tag{9}$$

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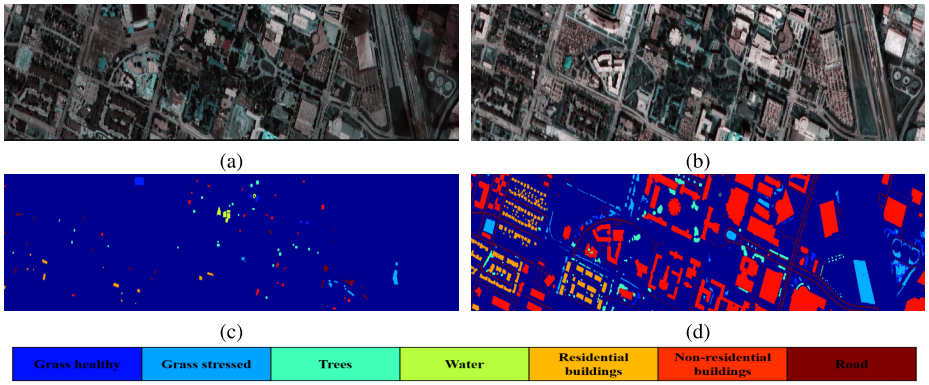


Fig. 4. Pseudocolor images and ground-truth maps of the Houston dataset. (a) Houston2013 pseudocolor image. (b) Houston2018 pseudocolor image. (c) Houston2013 ground-truth map. (d) Houston2018 ground-truth map.

where p_i^c and q_i^c represent the probability predictions from the CNN and GNN, respectively; n_s denotes the number of source samples; and C is the number of categories. The overall loss function is the combination of the classification loss, distribution alignment loss, graph alignment loss, and consistency loss, as detailed in the following:

$$L_{\text{Loss}} = L_{\text{STKD}} + L_{GOT} + L_{CE}^{CNN} + L_{CE}^{GNN} + L_{\text{Consistency}}. \quad (10)$$

After this step, both the SD and TD will exhibit more distinct feature information, mitigating the effects of domain shift and thereby enhancing the performance of cross-domain HSI classification.

IV. Experimental Results And Analysis

In Section IV, we offer a detailed description of the datasets and experimental setups that were part of our research. We then compare MKDnet with leading DA and domain generalization (DG) methods, showing that MKDnet outperforms these techniques. Lastly, we perform parameter experiments and conduct an ablation study on the essential modules of MKDnet.

A. Datasets

To assess the effectiveness of this approach, we experiment with six HSI datasets: Houston2013, Houston2018, Dioni, Loukia, Pavia University, and Pavia Center.

1) *Houston:* The Houston dataset consists of the Houston2013 and Houston2018 datasets. The Houston2013 dataset has a spatial resolution of 349 × 1905 pixels with a 2.5-m resolution, including 144 spectral bands and 15 classes. The Houston2018 dataset features 209 × 955 pixels with a higher resolution of 1 m, 48 spectral bands, and 20 classes. The datasets, collected in 2013 and 2018 with various sensors, encompass the University of Houston campus and nearby regions, demonstrating the domain shift due to sensor differences and temporal variations. Fig. 4 shows the pseudocolor images and ground-truth maps for the two datasets. The datasets span a wavelength range from 0.38 to 1.05 μm, and for the DA task, seven common classes were chosen, as indicated in Table I.

TABLE I
Number of Samples and Classes in the Source and Target Houston Dataset

Class		Number of Samples	
No.	Name	Houston2013 (Source)	Houston2018 (Target)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
Total		2530	53200

2) *HyRANK:* The dataset was created as part of the International Society for Photogrammetry and Remote Sensing (ISPRS) scientific initiative. It includes HSI captured by the Hyperion sensor (EO-1, USGS), with 176 spectral bands. The dataset includes two annotated scenes: Dioni, which has dimensions of 250 × 1376 pixels, and Loukia, with dimensions of 249 × 945 pixels. Both scenes contain 12 consistent classes. Notably, vegetation classification in these scenes is particularly prone to misclassification. Fig. 5 displays the pseudocolor images and ground-truth maps for Dinoi and Loukia. For the DA task, the analysis focused on the 12 common classes identified in these images, as shown in Table II.

3) *Pavia:* The dataset comprises the Pavia University and Pavia Center datasets. The Pavia University dataset originally contained 610 × 610 pixels and 115 spectral bands. The dataset was reduced to 103 bands, with 12 noisy bands removed, and further cropped to a size of 610 × 340 pixels after eliminating invalid data. The Pavia Center dataset initially had 1096 × 1096 pixels and 102 spectral bands, but it was resized to 1906 × 715 pixels due to invalid data. Both datasets were acquired by the ROSIS sensor over Pavia, Italy. Fig. 6 presents the pseudocolor images and ground-truth maps of both datasets, focusing on the seven consistent classes listed in Table III.

B. Experimental Setup

To explore the cross-domain HSI classification performance of MKDnet and evaluate it alongside seven other methods,

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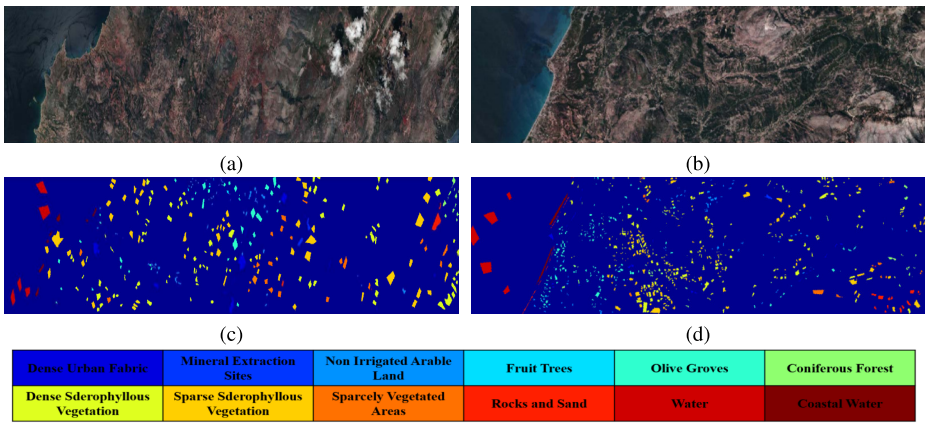


Fig. 5. Pseudocolor images and ground-truth maps of the HyRANK dataset. (a) Dioni pseudocolor image. (b) Loukia pseudocolor image. (c) Dioni ground-truth map. (d) Loukia ground-truth map.

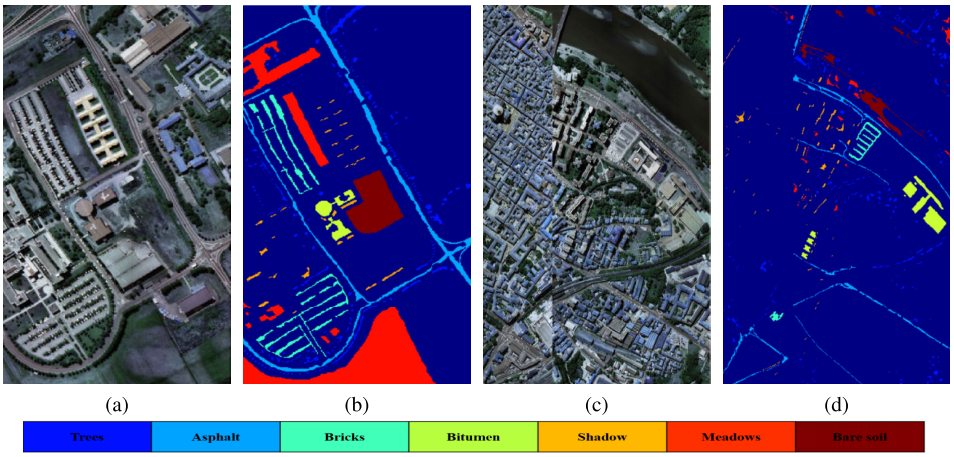


Fig. 6. Pseudocolor images and ground truth of the Pavia dataset. (a) Pavia University ground truth. (b) Pavia University pseudocolor image. (c) Pavia Center ground truth. (d) Pavia Center pseudocolor image.

TABLE II

NUMBER OF SAMPLES AND CLASSES IN THE SOURCE AND TARGET HYRANK DATASET

Class		Number of Samples	
No.	Name	Dioni (Source)	Loukia (Target)
1	Dense Urban Fabric	1262	206
2	Mineral Extraction Sites	204	54
3	Non Irrigated Arable Land	614	426
4	Fruit Trees	150	79
5	Olive Groves	1768	1107
6	Coniferous Forest	361	422
7	Dense Sderophyllous Vegetation	5035	2996
8	Sparse Sderophyllous Vegetation	6374	2361
9	Sparcely Vegetated Areas	1754	399
10	Rocks and Sand	492	453
11	Water	1612	1393
12	Coastal Water	398	421
Total		20024	10317

TABLE III

NUMBER OF SAMPLES AND CLASSES IN THE SOURCE AND TARGET PAVIA DATASET

Class		Number of Samples	
No.	Name	Pavia University (Source)	Pavia Center (Target)
1	Tree	3064	7598
2	Asphalt	6631	9248
3	Brick	3682	2685
4	Bitumen	1330	7287
5	Shadow	947	2863
6	Meadows	18649	3090
7	Bare soil	5029	6584
Total		39332	39355

DAN [31], MRAN [32], SagNet [57], TSTnet [33], DSAN [58], SDEnet [59], and S^{2}AMSnet [60]. Based on the analysis of the classification graphs, class-specific accuracy (CA), overall accuracy (OA), and the Kappa coefficient (KC) are employed as quantitative evaluation indicators for evaluating classification effectiveness. In terms of sample settings for model training, the DA-based methods DAN, MRAN, DSAN, TSTnet, and MKDnet select 80% of the labeled data from SD and all TD data for model training. For the DG-based SagNet, SDEnet, and S^{2}AMSnet, the model was trained using 80% of the SD-labeled data, with the remaining 20% allocated for

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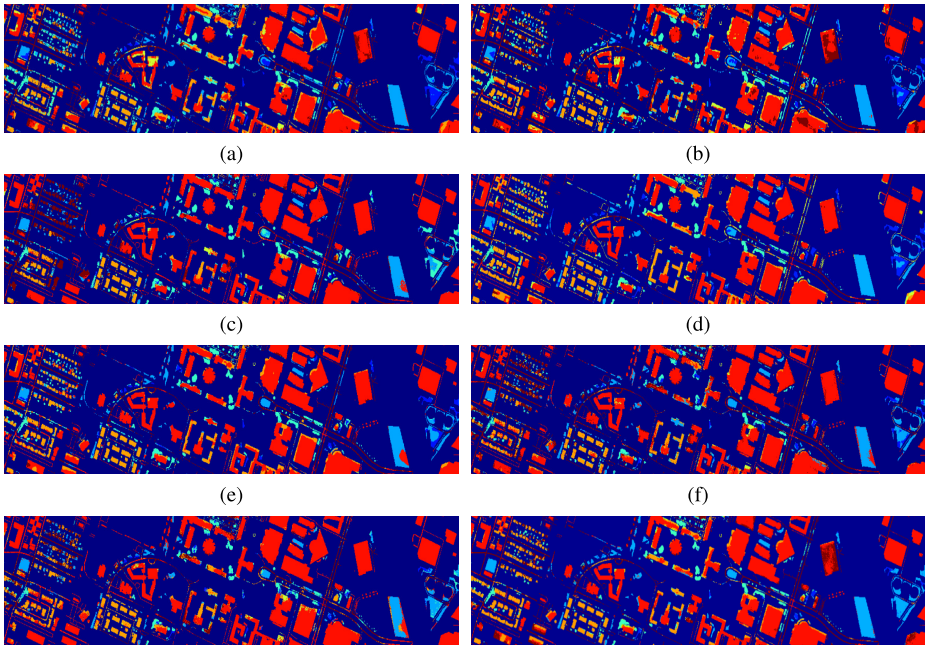


Fig. 7. Classification maps for the Houston2018 dataset generated with different methods. (a) DAN. (b) MRAN. (c) SagNet. (d) TSTnet. (e) DSAN. (f) SDEnet. (g) S^2AMSnet. (h) MKDnet.

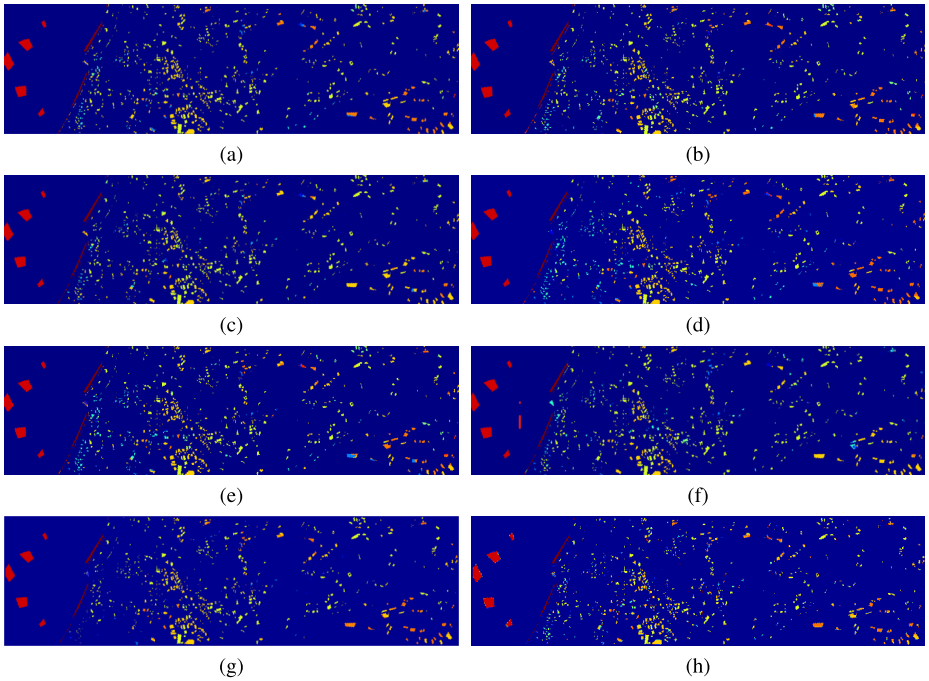


Fig. 8. Classification maps for the Loukia dataset generated with different methods. (a) DAN. (b) MRAN. (c) SagNet. (d) TSTnet. (e) DSAN. (f) SDEnet. (g) S^2AMSnet. (h) MKDnet.

validation. MKDnet is implemented and executed on the PyTorch platform, with computations performed on a 2080Ti GPU. The GraphSAGE model is implemented using the PyTorch Geometric library. The learning rate tuning strategy follows $\eta_w = (\eta_0/(1 + \alpha w)^\beta)$, where η_0 is the base learning rate, $\alpha = 10$, $\beta = 0.75$, and w represents the training progress, which varies linearly from 0 to 1. The total number of training iterations is set to 500. The optimization method uses a small batch stochastic gradient descent (SGD) algorithm with a momentum of 0.9.

C. Experiments and Results' Analysis

MKDnet shows outstanding performance in cross-domain HSI classification, as demonstrated by the classification maps presented in Figs. 7–9. These maps compare the results obtained by MKDnet with those of seven advanced methods

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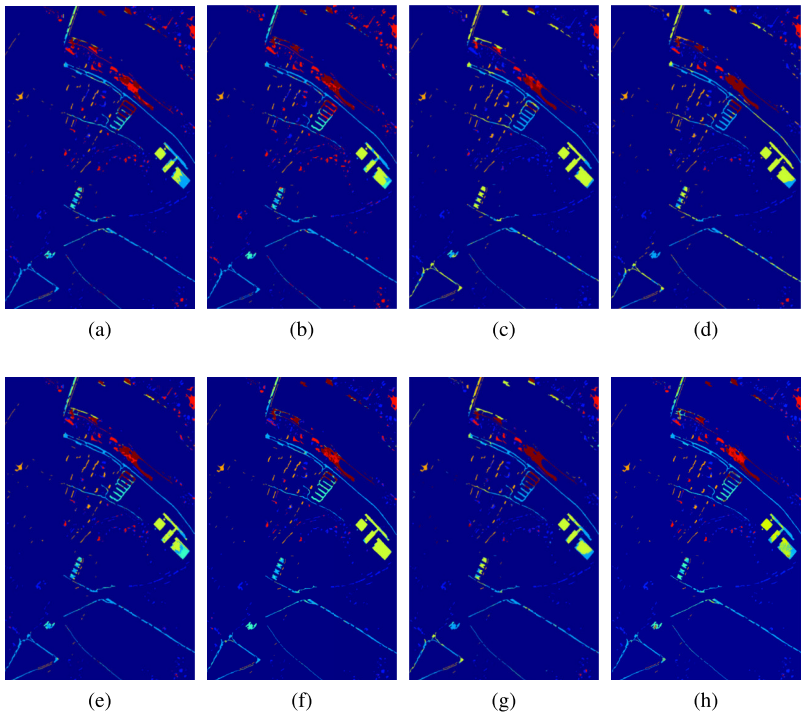


Fig. 9. Classification maps for the Pavia Center dataset generated with different methods. (a) DAN. (b) MRAN. (c) SagNet. (d) TSTnet. (e) DSAN. (f) SDEnet. (g) S^2AMSnet. (h) MKDnet.

TABLE IV
CLASSIFICATION RESULTS OF MKDNET AND OTHER METHODS ON THE TARGET SCENE HOUSTON2018 DATASET

Class	DAN	SagNet	MRAN	TSTNet	DSAN	SDEnet	S^2AMSnet	MKDnet
1	50.85	24.79	54.39	**82.03**	57.95	25.49	35.79	38.59
2	72.38	62.19	75.77	68.73	67.90	**83.26**	59.99	**85.29**
3	61.77	48.66	68.79	52.82	**71.69**	58.31	**77.64**	69.08
4	**100**	81.02	64.33	**100**	81.82	**100**	**100**	76.36
5	67.43	59.57	68.50	61.71	61.79	62.34	45.95	**75.58**
6	73.11	**88.29**	70.54	84.44	70.26	85.36	**90.12**	77.92
7	63.99	33.99	54.21	53.07	54.53	55.16	**64.48**	54.10
OA(%)	67.77	73.11	68.99	75.34	67.08	75.99	75.73	**76.11**
KC(×100)	51.17	55.39	52.67	59.69	52.05	60.77	60.54	**60.87**

across the Houston, HyRANK, and Pavia datasets. In the maps, unmarked pixels indicate the background, while marked pixels represent predicted classes, showing MKDnet's superior classification capability.

Existing methods have notable limitations. Feature extraction architectures such as DSAN, MRAN, and SagNet primarily focus on local information, limiting their ability to effectively handle long-range dependencies. MRAN's reliance on feature reuse strategies can lead to feature confusion, especially in cross-domain tasks where there are significant differences in feature spaces, negatively affecting the classification performance. DSAN may experience instability during its dynamic adaptation process, leading to performance fluctuations across different domains. On the other hand, S^2AMSnet and SDEnet prioritize global features but fail to capture important local details, resulting in poor performance in cross-domain HSI classification. In contrast, MKDnet integrates Mamba and CNN architectures, allowing it to extract both global and local features effectively. This approach enhances MKDnet's ability to handle long-range dependencies and provides more accurate classification in complex scenarios.

A detailed analysis of category-specific classification performance is provided in Tables IV–VI, which includes average accuracy and standard deviation across three target scenes. MKDnet consistently outperforms other methods in cross-domain HSI classification, with its OA surpassing DA methods like MRAN and DSAN by an impressive 1%–9% across critical target scenes, such as Houston2018, Loukia, and Pavia Center. Additionally, methods like DAN and TSTnet focus on marginal distribution alignment to reduce the distribution difference across different domains. However, they often overlook crucial feature differences between classes, which can negatively impact performance, especially when distinguishing similar classes. On the other hand, SagNet and

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TABLE V
CLASSIFICATION RESULTS OF MKDNET AND OTHER METHODS ON THE TARGET LOUKIA CENTER DATASET

Class	DAN	SagNet	MRAN	TSTNet	DSAN	SDEnet	S^2AMSNet	MKDnet
1	11.66	21.99	12.78	**39.81**	26.21	35.32	39.39	37.72
2	43.44	7.14	0	24.07	18.52	13.63	0	**46.11**
3	22.07	**49.70**	5.11	14.79	23.71	60.44	51.34	23.47
4	11.77	16.46	0	1.27	0	8.78	0	**25.70**
5	3.22	0.22	**28.43**	28.01	**48.69**	2.63	0	1.30
6	20.71	34.65	**40.51**	5.45	**45.97**	23.36	26.69	25.40
7	71.66	**80.77**	64.23	64.57	60.58	72.98	55.26	**73.47**
8	63.22	62.71	**70.99**	62.60	67.26	70.63	53.82	64.34
9	53.11	59.46	7.02	**73.68**	37.09	19.80	12.23	37.09
10	0	1.10	0.22	**2.21**	4.86	**18.77**	0	4.21
11	99.79	**100**	**100**	**100**	**100**	**100**	**100**	**100**
12	**100**	**100**	**100**	**100**	**100**	**100**	90.93	**100**
OA(%)	56.43	61.77	58.65	62.66	60.92	61.37	55.75	**63.01**
KC(×100)	44.69	52.12	49.32	54.20	52.44	52.99	45.78	**54.98**

TABLE VI
CLASSIFICATION RESULTS OF MKDNET AND OTHER METHODS ON THE TARGET PAVIA CENTER DATASET

Class	DAN	SagNet	MRAN	TSTNet	DSAN	SDEnet	S^2AMSNet	MKDnet
1	73.79	**96.34**	60.16	51.82	84.39	86.32	89.42	78.48
2	78.65	56.87	78.77	79.76	76.34	80.32	77.12	**81.96**
3	80.77	6.31	54.75	60.41	54.77	69.53	25.36	**51.10**
4	66.43	**79.66**	67.32	31.44	74.65	70.31	70.98	35.06
5	96.23	90.40	68.63	74.64	**98.66**	76.32	92.36	86.18
6	73.46	50.33	**86.45**	86.99	74.98	75.91	42.18	77.62
7	70.13	57.99	62.31	51.09	67.36	60.34	**67.62**	60.14
OA(%)	73.99	69.71	68.87	69.63	72.96	74.36	72.87	**74.94**
KC(×100)	68.89	63.14	62.36	64.08	68.32	69.12	67.65	**69.73**

TABLE VII
ABLATION STUDY OF MKDNET WITH DIFFERENT LOSS COMPONENTS ON OA (%) ACROSS THREE DATASETS

Dataset	w/o L_{STKD}	w/o L_{GOT}	w/o $L_{\text{CE}}^{\text{CNN}}$	w/o $L_{\text{CE}}^{\text{GNN}}$	w/o $L_{\text{consistency}}$	Full
Houston2018	73.53	75.36	2.54	74.11	76.01	**76.11**
Loukia	62.78	62.96	0.52	62.89	62.34	**63.01**
PaviaC	69.57	71.59	10.30	66.29	73.13	**74.94**

TABLE VIII
ABLATION ANALYSIS OF GLMC WITH DIFFERENT MODULE COMBINATIONS ON OA (%) ACROSS THREE DATASETS

No.	CNN	Mamba	Transformer	Houston2018	Loukia	PaviaC
1	✓	×	×	75.58	62.79	71.99
2	×	✓	×	75.73	62.85	72.03
3	✓	×	✓	75.93	62.89	72.87
4	✓	✓	×	**76.11**	**63.01**	**74.94**

SDEnet prioritize global alignment for DG but fail to capture local details, which are vital for fine-grained classification. Their emphasis on global alignment limits their ability to distinguish local structures, reducing accuracy in tasks that require precise class differentiation. In contrast, MKDnet adopts a more comprehensive approach. Its STKD module strengthens the relationships between classes, enhancing the extraction of both global and local features. The module enables the transfer of soft labels from the teacher network to the student network, facilitating feature transfer. This process effectively addresses domain shift and improves domain alignment. Moreover, through graph alignment, MKDnet captures intricate topological relationships, which further enhances its performance in cross-domain HSI classification.

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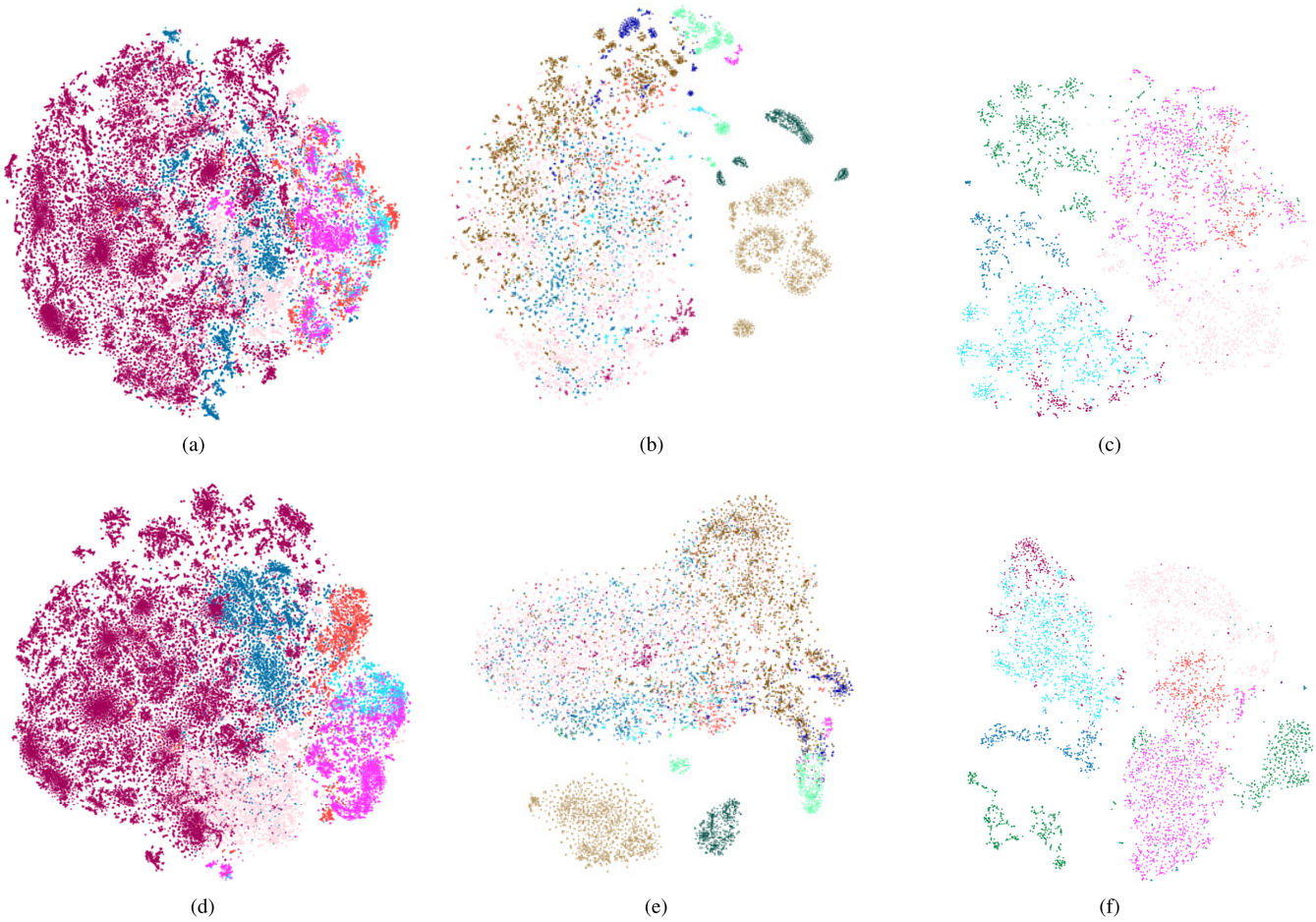


Fig. 10. Visualization of feature distributions before and after categorization by MKDnet. (a)–(c) Feature distributions before classification. (d)–(f) Results after classification.

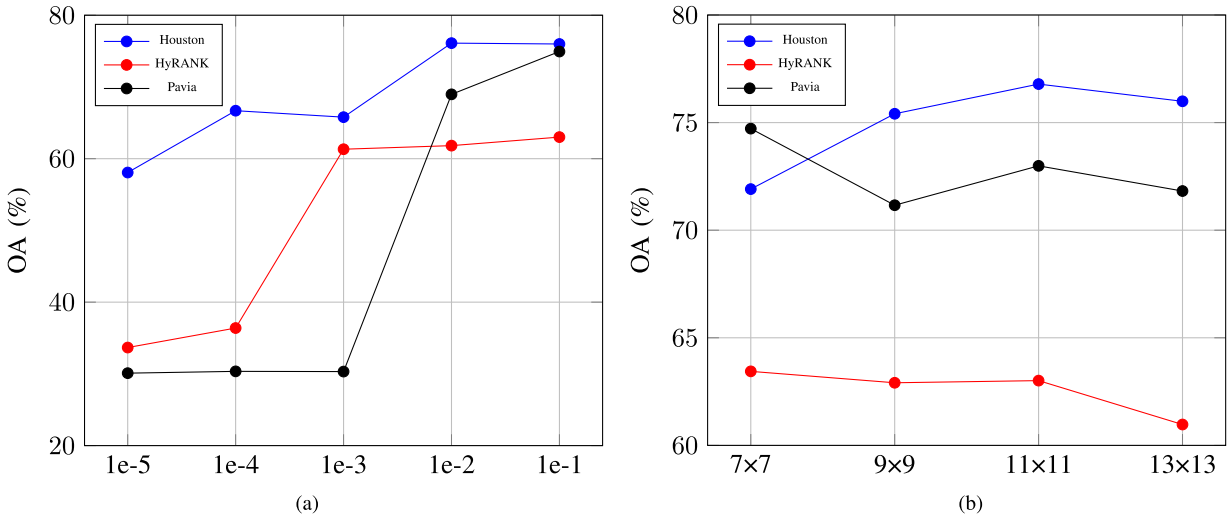


Fig. 11. Parameter tuning of different parameters. (a) Base learning rate. (b) Patch size.

In Fig. 10, we present the visualizations of the feature distributions before and after the classification process. It can be observed that the features before classification exhibit significant overlap and poor interclass separability, indicating limited discriminative power of the raw features. In contrast, after applying our proposed method, the feature representations become more compact within classes and more distinctly separated between classes, demonstrating that our model effectively enhances feature discriminability and thus improves classification performance.

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TABLE IX
ABLATION ANALYSIS OF MKDNET WITH DIFFERENT MODULE COMBINATIONS ON OA (%) ACROSS THREE DATASETS

No.	STKD	GLMC	Houston2018	Loukia	PaviaC
1	×	×	75.34	62.66	69.03
2	×	✓	75.98	62.98	72.69
3	✓	×	75.55	62.73	71.21
4	✓	✓	**76.11**	**63.01**	**74.94**

In conclusion, MKDnet significantly enhances cross-domain HSI classification accuracy through effective feature extraction and domain alignment techniques, providing an efficient solution to domain shift and feature discrepancy challenges.

D. Ablation Experiments

To evaluate the contribution of each loss component in MKDnet, we perform an ablation study by removing them individually. As shown in Table VII, removing $L_{\mathrm{CE}}^{\mathrm{CNN}}$ causes a drastic drop in performance on all datasets, demonstrating its fundamental role in model training. $L_{\mathrm{consistency}}$ also plays a critical role, especially on the PaviaC dataset, where its removal leads to a 1.81% decrease in accuracy. Additionally, L_{STKD}, L_{GOT}, and $L_{\mathrm{CE}}^{\mathrm{GNN}}$ contribute positively to improving classification accuracy. Overall, the full configuration achieves the highest accuracy, validating the effectiveness of combining all loss components.

We then conduct structural ablation experiments to further evaluate the effectiveness of the core modules in the proposed multilayer framework, particularly focusing on the GLMC and STKD modules, as shown in Tables VIII and IX.

As presented in Table VIII, we perform ablation experiments on the individual components of GLMC to further evaluate its effectiveness. According to the results, the CNN and Mamba combination achieves superior performance on the Houston2018, Loukia, and PaviaC datasets, outperforming the use of CNN or Mamba alone, as well as the CNN–Transformer combination. When using CNN alone, the model primarily focuses on local details in the hyperspectral data, while Mamba focuses on global information. The CNN–Transformer combination, although capable of extracting more information, comes with a higher computational complexity. In contrast, the Mamba–CNN combination not only extracts hyperspectral features more effectively but is also more efficient, leading to a significant enhancement in classification results.

To evaluate the impact of the STKD module, we maintain only the graph alignment module and leave other components as they are. The results demonstrate that the model performs significantly worse than the full MKDnet. This underperformance is primarily due to the graph alignment module focusing solely on topological relationships and neglecting interclass features, leading to a noticeable decline in overall performance.

Next, we replace the GLMC with VGG16 to assess the effectiveness of the GLMC module. This modification resulted in a 1%–2% drop in OA across all datasets. These findings underscore the GLMC module's advantage in feature extraction, as it not only captures local features efficiently but also integrates global information, which is crucial for improving cross-domain HSI classification.

The ablation experiment results demonstrate that the full MKDnet, comprising both the GLMC and STKD modules, achieves the highest OA across the three target scenes (Houston2018, Loukia, and PaviaC), as shown in Table IX. Notably, the model without the STKD module performed poorly on all datasets, indicating that the absence of this module hampers the model's capacity to comprehend interclass features, thereby diminishing classification performance. Similarly, replacing the GLMC module also led to a noticeable drop in OA, highlighting the negative impact of inadequate feature extraction on the model's accuracy. These results emphasize the critical role of the GLMC module in effectively capturing both global and local features.

Overall, the ablation experiments demonstrate that the combination of the STKD and GLMC modules is essential for effective cross-domain HSI classification, resulting in MKDnet's superior performance across all target scenes.

E. Parameter Tuning

In this section, we explore how parameter choices influence MKDnet's classification performance across three target scenes, focusing specifically on the base learning rate and patch size.

For the three cross-domain datasets, we determine the optimal base learning rate for MKDnet from a range of values: $1e^{-5}$, $1e^{-4}$, $1e^{-3}$, $1e^{-2}$, and $1e^{-1}$. As shown in Fig. 11(a), an increase in the base learning rate results in a rising trend in OA across all three datasets. Notably, for the Houston dataset, this improvement is relatively gradual, while for the HyRANK and Pavia datasets, there is a marked jump at specific learning rates, accompanied by greater fluctuations. The optimal base learning rates identified are $1e^{-2}$ for the Houston dataset and $1e^{-1}$ for both the HyRANK and Pavia datasets.In terms of patch size, we evaluated sizes of 7×7, 9×9, 11×11, and 13×13 across the three target scenes, with the results presented in Fig. 11(b). The analysis reveals that larger patch sizes do not consistently yield better performance. For the Houston dataset, performance improves with larger patch sizes, peaking at 11×11 before declining at 13×13. In contrast, the Pavia dataset shows a gradual performance decrease with larger patch sizes, while the HyRANK dataset maintains relatively stable performance regardless of patch size. This indicates that different scenes benefit from different patch sizes: the Houston dataset achieves optimal performance with a patch size of 11×11, whereas the HyRANK and Pavia datasets perform best with a patch size of 7×7.

F. Analysis of the Computational Complexity

In related research fields such as HSI classification, model training efficiency is crucial, and the duration of an epoch, i.e., the time required for the model to completely traverse the entire training dataset, is one of the key indicators of training efficiency. In this experiment, a single round training duration test is carried out on Houston, HyRANK, and Pavia datasets

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TABLE X
EXECUTION TIME FOR ONE EPOCH OF TRAINING (IN SECONDS) FOR DIFFERENT METHODS

	DAN	SagNet	MRAN	TSTnet	S^2AMSNet	DSAN	SDEnet	MKDnet
Houston2018	10.79	9.78	17.84	5.61	8.99	19.36	8.66	7.22
Loukia	16.72	18.66	21.46	30.82	26.31	15.73	18.56	36.61
PaviaC	30.11	22.48	28.14	7.21	22.79	26.76	22.44	8.36

for a variety of models such as DAN, SagNet, MRAN, TSTnet, S^2AMSNet, DSAN, SDEnet, and MKDnet, and the specific results are shown in Table X.

MKDnet demonstrates notable training efficiency in HSI classification. On the Houston2018 and PaviaC datasets, its one epoch training time is significantly lower than most compared methods, achieving up to 63% speed improvement over DSAN and SagNet, ranking as the fastest or second-fastest overall. This efficiency benefits from the synergy of Mamba's global modeling capability, CNN's local feature extraction, and a lightweight knowledge distillation design. Although training time increases on the Loukia dataset, possibly due to complex spatial structures or distillation overhead, MKDnet still achieves a favorable balance between speed and performance across all three datasets.

V. CONCLUSION

In cross-domain HSI classification, extracting comprehensive features and reducing domain shift are significant challenges. In this article, we design MKDnet to address the above issues. A GLMC module is used to extract both global and local features. The effective domain alignment includes both distribution alignment and graph alignment. The distribution alignment module employs knowledge distillation techniques to reduce model complexity, enhance the understanding of interclass features, and mitigate the effects of domain shift. For graph alignment, graph convolution techniques are employed to capture the relationships between nodes and edges, thereby enhancing the understanding of topological relationships. Finally, consistency constraints are applied to both distribution alignment and graph alignment, and classification is performed based on their outputs, ensuring better feature balance across different classes. Extensive experiments on three cross-domain HSI datasets demonstrate that MKDnet successfully reduces domain shift and enhances classification accuracy.

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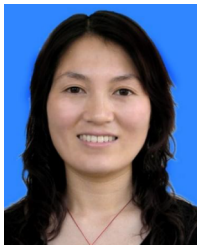
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Aoyan Du received the B.E. degree in computer science and technology from Qilu Normal University, Jinan, China, in 2023. She is currently pursuing the M.E. degree with the School of Computer Science and Technology, Qilu University of Technology (Shandong Academy of Sciences), Jinan.

Her research interests include computer vision and image processing.



Guixin Zhao received the Ph.D. degree from the School of Information and Control Engineering, China University of Mining and Technology, Xuzhou, China, in 2021.

From 2016 to 2017, she was a Visiting Scholar of Brunel University London, London, U.K. She is currently a Lecturer with the Faculty of Computer Science and Technology, Qilu University of Technology (Shandong Academy of Sciences), Jinan, China. Her research interests mainly include machine learning and computer vision.

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Mengxin Cao (Member, IEEE) received the B.B.M. degree from the School of Information Management and Information System, Shandong First Medical University, Jinan, China, in 2021. She is currently pursuing the master's degree with the Faculty of Computer Science and Technology, Qilu University of Technology (Shandong Academy of Sciences), Jinan.

She has authored or co-authored two papers in IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS) and the IEEE International Conference on Image Processing (ICIP). Her research interests include computer vision and image processing.



Yuyao Wang received the B.S. degree in mathematics from Shandong University, Jinan, China, in 2019, and the M.A. degree in statistics from Columbia University, New York, NY, USA, in 2020. She is currently pursuing the Ph.D. degree in statistics with Boston University, Boston, MA, USA.

She was a part of the Hua Loo-Keng Talent Program at Shandong University. Her research interests include transfer learning, causal inference, nonparametric statistics, and generative models.



Aimei Dong received the B.S. degree in computer science from the Ocean University of China, Qingdao, China, in 2001, and the Ph.D. degree in information technology and engineering from Jiangnan University, Wuxi, China, in 2016.

She is currently an Associate Professor with the Faculty of Computer Science and Technology, Qilu University of Technology (Shandong Academy of Sciences), Jinan, China. She visited the University of North Carolina at Chapel Hill, NC, USA, from January 2018 to January 2019. She has authored or co-authored over 30 research articles in international/national journals. Her research interests include subspace analysis, medical image analysis, deep learning for multimodality data, semi-supervised learning, and transfer learning.



Guohua Lv received the Ph.D. degree in computer science from Monash University, Clayton, VIC, Australia, in 2015.

He is currently an Associate Professor with the Faculty of Computer Science and Technology, Qilu University of Technology (Shandong Academy of Sciences), Jinan, China, where he is the Leader of the Vision and Information Processing Laboratory. He has published over 30 refereed journals and conference papers. His research interests mainly include computer vision and machine learning.

Dr. Lv is a reviewer for some international journals, such as IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS, IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, and *Computer Vision and Image Understanding and Neurocomputing*.



Yongbiao Gao received the B.Sc. degree in computer science and technology from the University of Jinan, Jinan, China, in 2015, the M.Sc. degree in computer technology from China University of Petroleum (East China), Qingdao, China, in 2018, and the Ph.D. degree in software engineering from Southeast University, Nanjing, China, in 2023.

He is currently a Lecturer with the School of Computer Science and Technology, Qilu University of Technology (Shandong Academy of Sciences), Jinan. He has published many academic papers, including IJCAI, IEEE Transactions on Neural Networks and Learning Systems (TNNLS), and ICME. His research interests include label distribution learning, reinforcement learning, and computer vision.



Dongqing Li received the Ph.D. degree in control science and engineering from the School of Information and Control Engineering, China University of Mining and Technology, Xuzhou, China, in 2014.

He is a currently a Lecturer with the Center of Software and Simulation, System Engineering Research Institute, Beijing, China. His research interests include machine learning, hyperspectral data processing, and software simulation.



Xiangjun Dong received the Ph.D. degree in computer applications from Beijing Institute of Technology, Beijing, China, in 2005.

From 2007 to 2009, he worked as a Post-Doctoral Fellow with the School of Management and Economics, Beijing Institute of Technology. From 2009 to 2010, he was a Visiting Scholar at the University of Technology Sydney, Sydney, NSW, Australia. He is currently a Professor with the School of Computer Science and Technology, Qilu University of Technology (Shandong Academy of Sciences), Jinan, China. He has authored or co-authored more than 100 journal/conference publications, including *Artificial Intelligence*, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON CYBERNETICS, *Pattern Recognition*, and the Conference on Information and Knowledge Management (CIKM). His research interests include data mining, artificial intelligence, and big data.

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